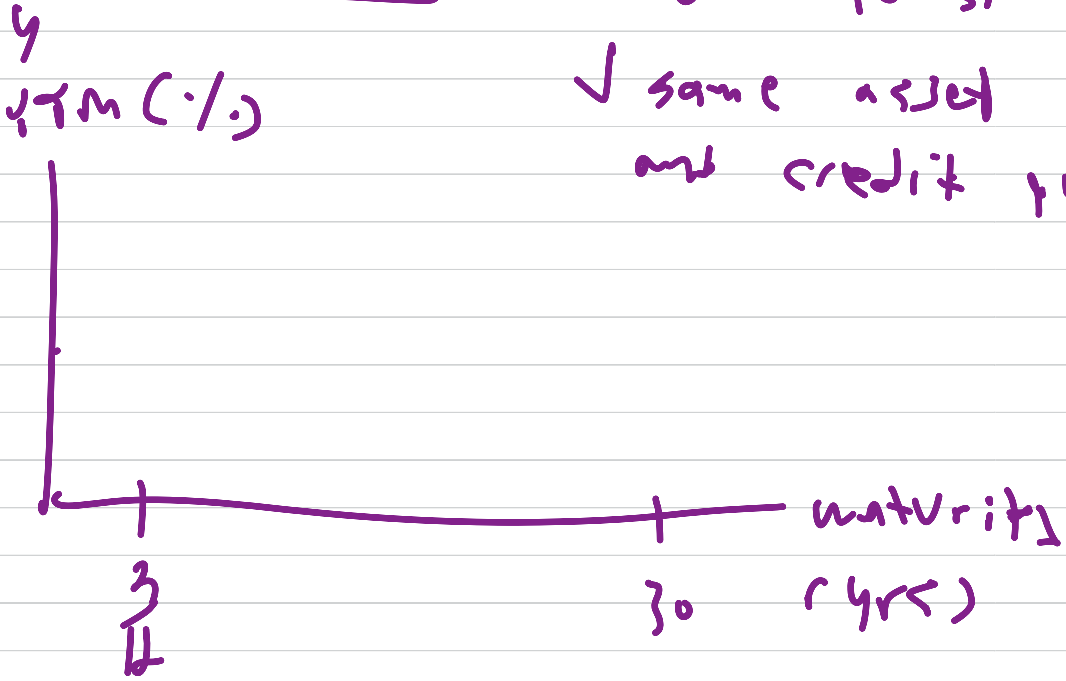
$\boxed{\text{Yeild Curve}}$ — 'term structure of interest rate'

✓ same asset class and credit quality



Current yield = $\dfrac{\text{Annual Coupon}}{\text{Purchase Price}}$

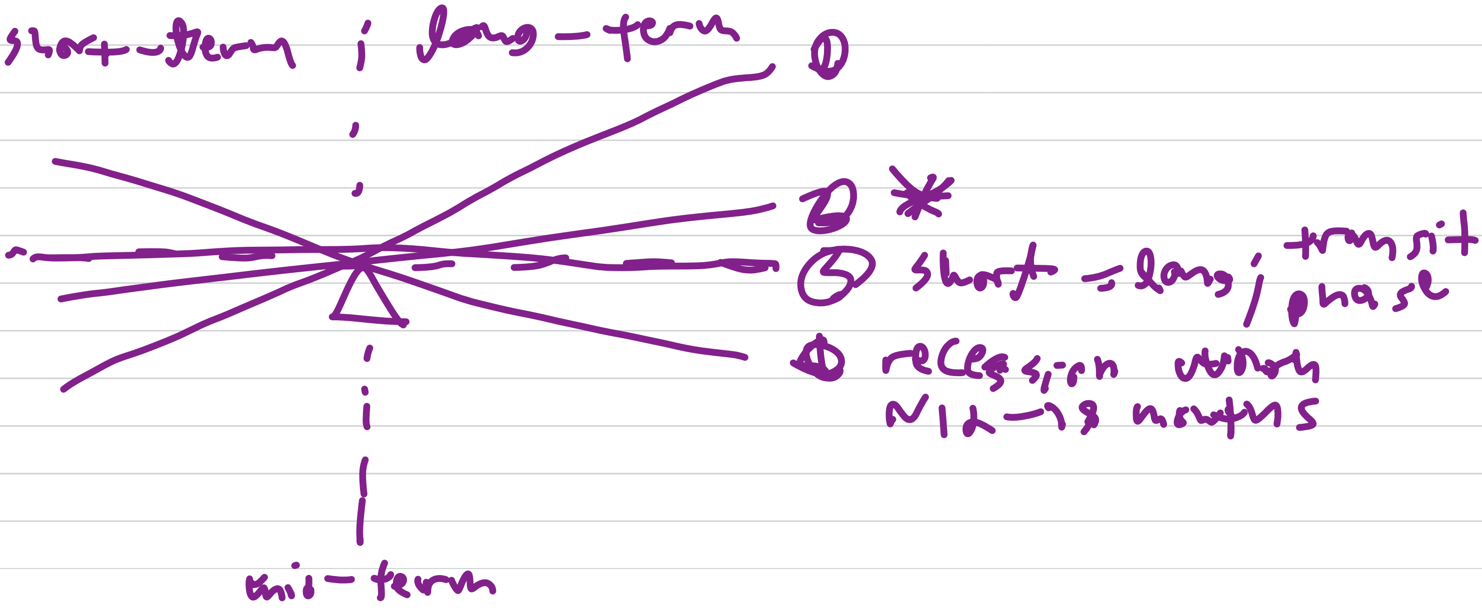
YTM = total return

+ appre./decre. + interest-on-interest

good for vs.

diff. characteristics

short-term ; long-term



Theory

short-term → Pure $\mathbb{E}(\cdot)$
'slope reflect only $\mathbb{E}(\cdot)$ of future short term rate'

long-term → Liquidity preference
'LT yield include term (liquidity) premium on top of rate expect'

maturity → Preferred habitat
'investor prefer maturity habitat = require premium to move them'

Factor

1. level → all maturity move ≈, 'parallel'
2. slope → ST > LT rate
3. curvature → MT change most

Drivers

1. ST → FED fund rate ~80-90% explained
'control short-end'
2. LT → inflation $\mathbb{E}(\cdot)$
drive level factor
~5yr avg. core CPI ~66%

③ Real Macro. Activity
↓ MT
↓ LT via level factor

LT yield = expect component
(future st rate)
+
risk (term) premium
(compensation for
uncertainty)

Standard Model → ~~x~~ overest. rate risk
premium reduction
vs
Bias-corrected Model : risk premium → counter cyclical

MVP: 10y − 3m spread → capture
full year-term
expectation

App 1. leading indicator
2. benchmark pricing
3. portfolio strategy → earn above
avg. bond returns

Liquidity

Ratio	Formula
Current Ratio	CA ÷ CL
Quick Ratio	(CA − Inventory − Prepaid) ÷ CL
Cash Ratio	Cash ÷ CL
Net Working Capital	CA − CL

Solvency

Ratio	Formula
Debt-to-Equity	Total Debt ÷ Equity
Debt Ratio	Total Liabilities ÷ Total Assets
Equity Multiplier	Total Assets ÷ Equity
Interest Coverage	EBIT ÷ Interest Expense
DSCR	NOI ÷ (Principal + Interest)

Profitability

Ratio	Formula
Gross Margin	Gross Profit ÷ Revenue
Operating Margin	EBIT ÷ Revenue
Net Margin	Net Income ÷ Revenue
ROA	Net Income ÷ Avg. Total Assets
ROE	Net Income ÷ Equity
ROIC	NOPAT ÷ Invested Capital
DuPont ROE	Net Margin × Asset Turnover × Equity Multiplier

Efficiency

Ratio	Formula
Asset Turnover	Revenue ÷ Avg. Total Assets
Inventory Turnover	COGS ÷ Avg. Inventory

Ratio	Formula
DIO	365 ÷ Inventory Turnover
AR Turnover	Revenue ÷ Avg. AR
DSO	365 ÷ AR Turnover
AP Turnover	COGS ÷ Avg. AP
DPO	365 ÷ AP Turnover
CCC	**DIO + DSO − DPO**

Market Valuation

Ratio	Formula
EPS	(Net Income − Preferred Div.) ÷ Shares Outstanding
P/E	Market Price ÷ EPS
P/B	Market Price ÷ Book Value per Share
EV	Mkt Cap + Total Debt − Cash
EV/EBITDA	EV ÷ EBITDA
Dividend Yield	DPS ÷ Market Price
Payout Ratio	Dividends ÷ Net Income